ARMSYNC | ROBOTIC ARM HACKATHON

PARTICIPANT BRIEF

Mode-Based Web-Controlled Robotic Arm • 48-Hour Hackathon

The challenge

ArmSync challenges you to build a robotic arm operated through a web server. Your solution must bring together embedded control, sensing, computer vision where applicable, and web development. All three tiers must be presented on one website as separate tabs. Switching tabs changes the arm's active operating mode.

Objective: Develop a mode-based, web-controlled robotic arm that can be driven manually with live video (Tier 1A), pick and place an object detected by a distance sensor (Tier 1B), and classify and sort a cube, a sphere and a pyramid into their respective boxes using computer vision (Tier 1C).

Challenge tiers

Tier	Mode	Expected behaviour
1A	Manual Control	A website provides sliders or buttons to move each joint and open or close the gripper. Show a live ESP32-CAM feed beside the controls. Enforce joint limits and smooth, rate-limited motion. Show connection status and current joint angles.
1B	Sense & Place	A distance sensor detects an object at a fixed pickup point. The arm automatically picks it up and places it at one predefined location. The dashboard displays live distance, task state and pick count.
1C	Vision Sorting	Computer vision detects and classifies a cube, sphere and pyramid. The arm sorts each shape into its respective box. The dashboard displays the annotated camera view, predicted class and sort log.

One website, three modes

Use one website with a tab for each mode (1A Manual, 1B Sense & Place, 1C Vision Sort). Switching tabs must stop the previous routine, hold the arm's position and then give control to the newly selected mode. Only one mode may command the arm at a time. A stop button must remain visible on every tab.

What you need to build

- A working robotic arm with controlled joints and a gripper.
- A web server and tabbed dashboard that can send commands to the arm and show relevant status, sensor readings and results.
- Tier 1A: manual joint/gripper controls and a live ESP32-CAM feed.
- Tier 1B: distance-sensor detection and automatic pick-and-place at a predefined location.
- Tier 1C: computer-vision classification of cube, sphere and pyramid, followed by sorting into the correct boxes.
- Safe motion handling, including joint limits, a stop control and a safe response to a lost connection.

How your work will be evaluated

Judges will assess demonstrated functionality, safe and reliable arm control, web integration, sensing and vision performance, testing, and your understanding of the implementation. Tier 1A and Tier 1B do not require machine learning. Tier 1C must classify all three specified shapes and demonstrate sorting.

Evaluation evidence to prepare

- A live demonstration of each completed tier.
- The website and its three mode tabs, including safe mode switching and the always-visible stop control.
- Source code and a short explanation of the hardware, software, communication flow and key design decisions.
- For Tier 1C, held-out test results such as per-class precision/recall or a confusion matrix, plus real sorting trials.
- Clear identification of any simulated, prerecorded or missing data wherever used.